



kaggle

Kaggle 5th Presentation

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Agenda

1. Background
2. Summary
3. Key Processing
4. Training methods
5. Important findings
6. Simple model

Background

Professional/Academic Background

- Third-year undergraduate at Chongqing University of Posts and Telecommunications (CQUPT).
- Interested in Machine Learning, Deep Learning, and Computer Vision.

Prior experience

- Participated in data science competitions on Kaggle and Alibaba Tianchi.
- Gained hands-on experience in:
 - Data Preprocessing
 - Model Training & Augmentation
 - Model Ensemble

Training methods

- CNN backbone & SED head
- Perch v2 embedding Aux-distillation
- Distillation
- Data Augmentation

Key Performance Contributors

- Pretrained backbone
- Perch v2 embeddings
- TTA

Training time

- About 60s-180s per epoch

Key Processing

Single Model Processing

Data Aug

- MixUp & SpecAugment & FilterAugment & Wave-noisy

Pretrained backbone init

- 2025 BirdCLEF 2nd place

TTA

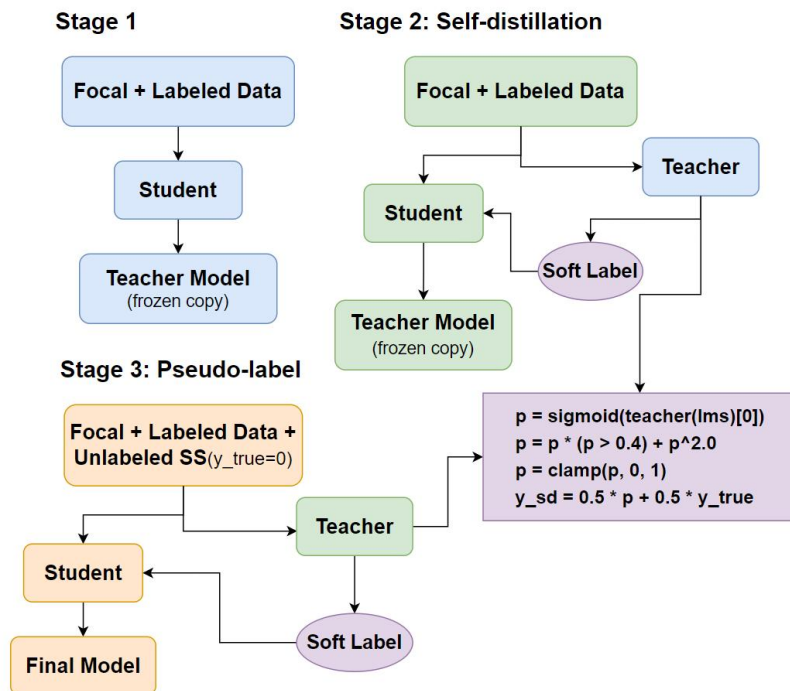
- 5s base window + ± 2.5 s shifted crops

Post-Processing

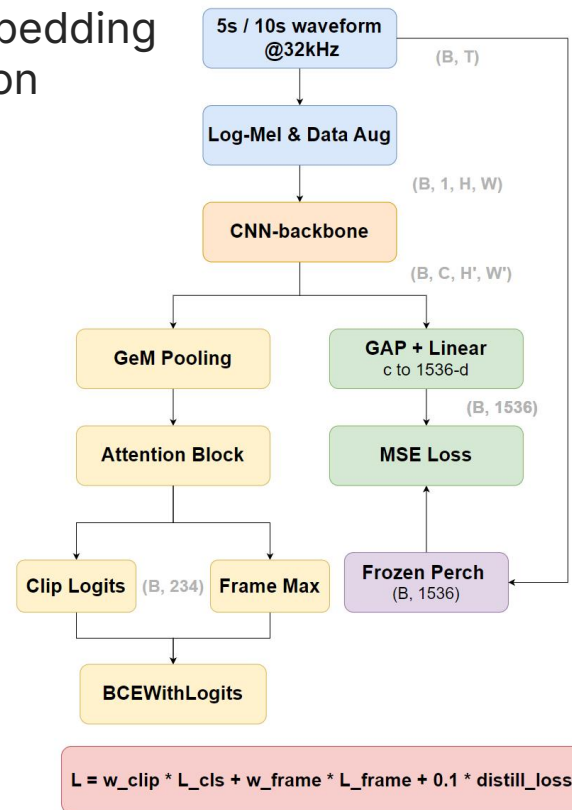
- file_peak_scale
- temporal_smooth_on_probs

Improvement	Public Score
Baseline	0.88–0.895
+ Data Aug	0.895–0.905
+ Perch-Distilled	0.905–0.915
+ Pretrained	0.915–0.925
+ Self-Distilled	0.925–0.928
+ Pseudo iter 1	0.928–0.935
+ TTA	+0.005–0.008
Post-Processing	+0.005–0.01

Self-Distilled & Pseudo unlabeled



Perch v2 embedding Aux-distillation



Training Methods

Training methods

backbone	pretrain	lms_shape	duration	loss	rare-class upsampling (66)	optimizer	scheduler	warm up epoch	total_epoch	lr
hgnetv2b0	-	128, 313	5s	bce	√	AdamW	LinearLR_warmup CosineAnnealingLR	5	35	8e-4
efficientnet v2s	bird-clef-2025-all-pretrained-models	128, 313	5s	bce	×	AdamW	OneCycleLR	5	30	4e-4
efficientnet v2s	bird-clef-2025-all-pretrained-models	128, 626	10s	bce	×	AdamW	OneCycleLR	5	30	4e-4
efficientnet b3	-	384, 384	5s	bce	×	AdamW	OneCycleLR	5	30	1e-3

Ensemble

- My CNN branch:
HGNetV2-B0, EfficientNetV2-S (5s), EfficientNetV2-S (10s), EfficientNet-B3
- Public branch: ProtoSSM, Distilled SED

$$\text{Final} = 0.6 \times (0.25 \cdot \text{hgnet} + 0.40 \cdot \text{effv2_5s} + 0.25 \cdot \text{effv2_10s} + 0.10 \cdot \text{effb3}) + 0.4 \times (0.6 \cdot \text{proto} + 0.4 \cdot \text{sed})$$

Important and Interesting Findings

Perch embedding distillation

- Used as an auxiliary loss
- Not the main classification objective
- Helped regularize training
- Improved convergence speed and stability

FilterAugment bug that worked

- An accidental implementation error remained unnoticed until after the competition
- Ablation showed that removing it decreased the score by 0.01+
- Although it was a bug, it behaved like a strong frequency-domain augmentation

```
# X: (B, F, T)
```

```
freq_filt = build_pieewise_linear_db_curve(F)
```

```
X_aug = X * 10 ** (freq_filt / 10)
```

Simple Model

Simple Model

- Main backbone: **EfficientNetV2-S 5s**
- Key features to keep: log-mel spectrogram, Perch embedding distillation, MixUp, FilterAugment, self-distillation, pseudo unlabeled ss
- Estimated private score: roughly 0.945+

Why this subset: captures most of the performance gain while staying much simpler than the full ensemble

Thank you